AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2018

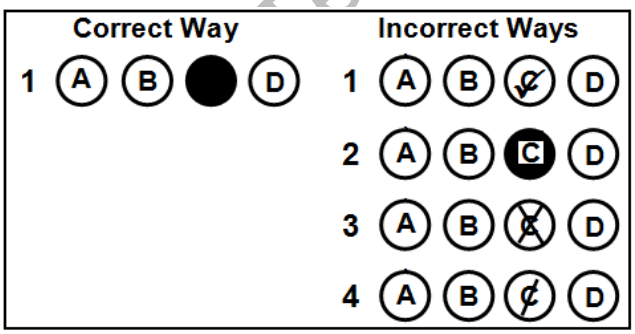
Physics Paper I

Time: 35 minutes Marks: 25

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. If the particles of a medium move back and forth along the direction of propagation of a wave, then this type of wave is known as

   A. surface wave.
   B. transverse wave.
   C. longitudinal wave.
   D. electromagnetic wave.

2. If a mechanical wave moves from a lighter to a denser medium, then which of the following factors will change during refraction?

   I. Wavelength
   II. Speed of wave
   III. Frequency of wave

   A. I only
   B. III only
   C. I and II
   D. II and III

3. The characteristic of sound by which two sound waves of same loudness and pitch can be distinguished from each other is called the

   A. speed of sound.
   B. quality of sound.
   C. pressure of sound.
   D. intensity of sound.

4. The amount of sound energy transmitted per second through unit area which is held perpendicularly in the direction of sound waves is called the

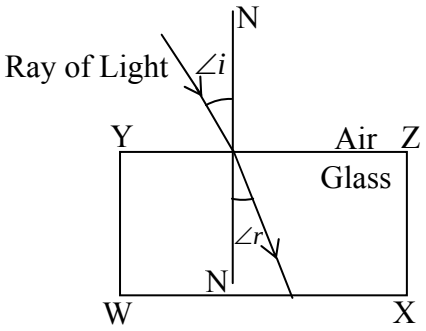
   A. pitch of sound.
   B. intensity of sound.
   C. loudness of sound.
   D. frequency of sound.

5. In the given figure, if the angle of incidence $(\angle i)$ is 40°, then the angle of refraction $(\angle r)$ will be

   (**Note**: The refractive index of glass is 1.5)

   I. less than 40°
   II. equal to 40°
   III. more than 40°

   A. I only.
   B. II only.
   C. I and III.
   D. II and III.



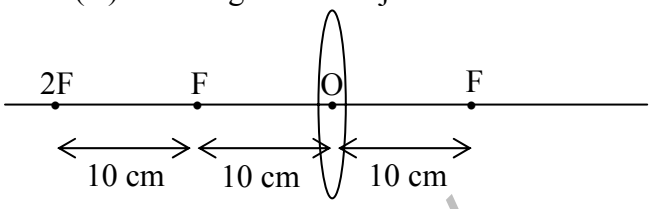
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6. When a white light passes through a prism it bends into seven different colours. This bending of colours increases as the

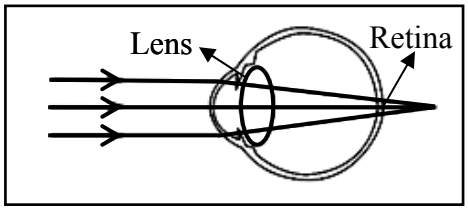
   A. intensity of light increases.
   B. amplitude of light increases.
   C. frequency of light decreases.
   D. wavelength of light decreases.

7. In the given ray diagram, if an object is placed at (2F) in front of a converging lens, then what will be the distance between the optical centre (O) and image of the object?

   A. 8 cm
   B. 12 cm
   C. 20 cm
   D. 36 cm



8. The given diagram shows the position of an image produced by an eye of a very distant object.



Which of the following correctly indicates the defect of vision and its correction measure as shown in the diagram?

	Defect of Vision	**Correction Measure**
A	Long sightedness	Use of a convex lens
B	Long sightedness	Use of a concave lens
C	Short sightedness	Use of a convex lens
D	Short sightedness	Use of a concave lens

9. Which of the following statements is FALSE about electric field lines?

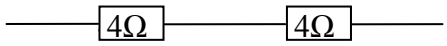
   A. They cross each other.
   B. They are imaginary lines.
   C. They are vector quantities.
   D. They move towards negative charge.

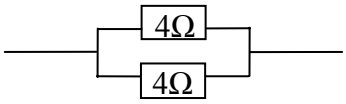
10. A capacitor is a device which

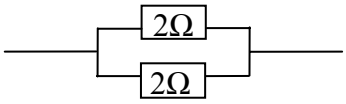
   A. stores electric charges.
   B. allows flow of direct current.
   C. blocks flow of alternating current.
   D. converts alternating current to direct current.

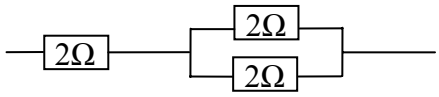
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11. Which of the following options gives the equivalent resistance of 2 Ω?

A. 

B. 

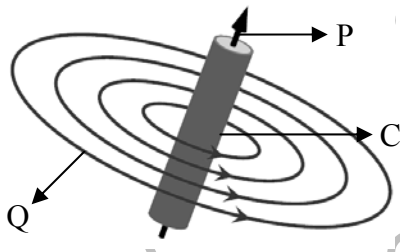
C. 

D. 

12. The quantity that remains the same in parallel combination of resistance is

A. voltage.
B. conductance.
C. electric power.
D. electric current.

13. In the given diagram of the current carrying conductor, the labelled points (P) and (Q) represent



	Point (P)	Point (Q)
A	Direction of Current	Electric Field
B	Direction of Current	Magnetic Field
C	Magnetic Field	Direction of Current
D	Electric Field	Direction of Current

14. A transformer is an electrical device that works on the principle of

A. self-induction.
B. Coulomb's law.
C. mutual induction.
D. Fleming's left hand rule.

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15. The energy conversion that takes place in the alternating current generator is from

    A. electrical to kinetic.
    B. mechanical to potential.
    C. mechanical to electrical.
    D. electrical to mechanical.

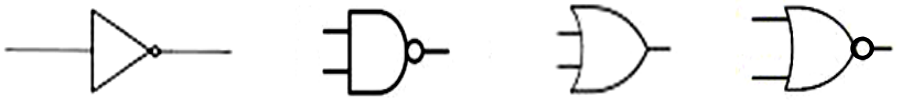
16. The magnitude of an induced electromotive force is NOT affected by the

    A. resistance in the coil.
    B. number of turns in the coil.
    C. amount of current passing through the coil.
    D. speed of relative motion between the coil and the magnet.

17. In thermionic emission, when a filament is heated, a beam of tiny particles is emitted from it. These particles are known as

    A. protons.
    B. electrons.
    C. beta particles.
    D. alpha particles.

18. The symbol that represents NAND gate is



    A B C D

19. When radiowaves are received by an antenna, the antenna converts the waves into

    I. electrical signals.
    II. electronic signals.
    III. electromagnetic signals.

    A. I only
    B. III only
    C. I and II
    D. II and III

20. If a call is made from mobile/ cell phone (A) to mobile/ cell phone (B), then radio signals of a particular frequency are first sent to the

    A. mobile/ cell phone (B).
    B. mobile switching centre.
    C. base station of the caller.
    D. base station of the receiver.

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21. Which of the following is a negatively charged particle?

A. Beta
B. Alpha
C. Proton
D. Neutron

22. Which of the following has the highest ionising and penetrating power?

	Ionising Power	**Penetrating Power**
A	Alpha particle	Beta particle
B	Alpha particle	Gamma rays
C	Beta particle	Alpha particle
D	Beta particle	Gamma rays

23. Carbon dating is a method for determining the age of an object containing organic material.

Which of the following carbon isotopes is used for this purpose?

A. Carbon-12
B. Carbon-13
C. Carbon-14
D. Carbon-15

24. During nuclear fusion reaction, when the hydrogen nuclei are converted into helium atom

A. less amount of energy is released.
B. huge amount of energy is absorbed.
C. amount of reactants decrease due to the energy released.
D. amount of reactants increase due to the energy absorbed.

25. All of the following are harmful effects of nuclear radiations on human beings EXCEPT that it causes

A. enlargement of the liver.
B. cancer of the blood cells.
C. redness and sores on the skin.
D. formation of cataract in the eye.

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